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Geometrical pattern effect on silicon deep etching by an inductively coupled plasma system

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Abstract
The etching rate in silicon deep reactive ion etching (RIE) is related to pattern geometry and a frequently seen defect, RIE lag, appears in feature sizes up to hundreds of micrometers. Different feature dimensions of rectangles, squares and circles/doughnuts are designed to realize how the geometrical pattern affects RIE lag in the inductively coupled plasma (ICP) etching process. Experimental results reveal that the primary dominating factor in RIE lag is feature width and secondary factors are feature area, shape and length-to-width ratio. Etching rates of rectangular trenches are sensitive to width while ring trenches are sensitive to both width and area. Process parameters are also adjusted to control RIE lag magnitude and realize its mechanism. The inverse RIE lag phenomenon appears at a much higher pressure of APC (auto pressure control) 75% at constant area features. The formation and removal of passivation film at the trench bottom will delay Si etching by F radical density, which will start earlier in a small width than a large one. It will be more obvious at higher pressure and lead to the reduction of RIE lag. This indicates that the cause of RIE lag in ICP etching is primarily attributed to the formation and removal of passivation film at the bottom of the trench, together with feature geometry. The RIE lag-eliminated trenches with constant area are obtained at a higher pressure of APC 70%. Deep and high aspect ratio silicon microstructures can be controlled by ICP etching with different pattern geometry.

1. Introduction

Silicon deep reactive ion etching (RIE) for deep and high aspect ratio microstructures is a very important technology in the microfabrication and microelectromechanical systems (MEMS) industry [1–6]. High density, low pressure plasma etching systems such as the inductively coupled plasma (ICP) system [7–9] have been developed to meet this requirement. The etching rate is related to the pattern geometry and the process parameters, e.g. coil power, bias, gas pressure, gas flow and the switching time at etching and passivation steps. RIE lag in ICP etching is a frequently seen defect in semiconductor or microfabrication processes and appears in MEMS feature sizes up to hundreds of micrometers. It will affect the etching micro-uniformity and is highly dependent on the pattern geometry. The effect is more severe as the feature width becomes smaller. Jiao *et al* [10] proposed that the etching rate under the silicon deep etching process is much affected by the mask layout related area, including the other features around it. Kiihamaki and Franssila [11] reported that the RIE lag is related to the feature length-to-width ratio (L/W ratio), width and pattern shape. Jansen *et al* [12] reported that RIE lag could be improved by lower pressure. Ayon *et al* [13] proposed that high SF_6 flow rate, the dominant variable, can minimize RIE lag. It would be interesting to understand the dominating factor of pattern geometry affecting RIE lag in ICP etching and find a better approach for RIE lag control. It would be very helpful for the process control of etching uniformity or applications using RIE lag for different levels or curved surface devices [14].

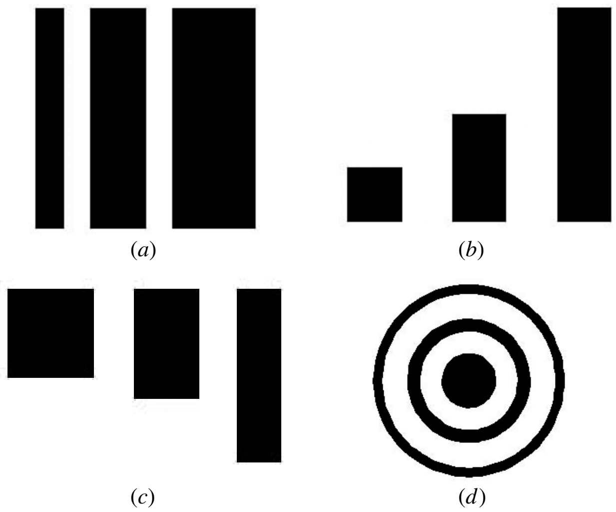


Figure 1. The schematic mask pattern for the study of geometrical pattern effect on RIE lag: (*a*) constant length, (*b*) constant width, (*c*) constant area of rectangles and (*d*) constant area of circles/doughnuts with different feature sizes.

In this paper, different shapes of rectangles, squares and circles/doughnuts were designed in three major kinds of patterns at constant length, constant width and constant area to study the pattern geometry effect on RIE lag in ICP etching. Process parameters were also applied to control RIE lag magnitude and realize its mechanism. Deep and high aspect ratio silicon microstructures were also demonstrated.

2. Experimental procedure

Boron-doped p-Si(100) wafers of 4–10 Ω cm were initially cleaned in H_2SO_4:H_2O_2 (3:1) solution. Standard lithography was used to transfer the photo mask pattern to the silicon surface and the etching mask was made photoresistant. The different feature geometry of rectangles, squares, circles and doughnuts is designed to realize how the pattern geometry dimension affects the ICP etching properties. Three major kinds of patterns of constant length, constant width and constant area with different feature sizes of 2 to 100 μm are used to understand the dominating factor of geometry for RIE lag in ICP etching as shown in figures 1(*a*)–(*d*). Silicon DRIE was performed by the ASE^{TM} process in an STS Multiplex ICP system [8, 9]. The etching and passivation gases were SF_6 and C_4F_8 respectively and switched during the process. A small amount of O_2 was added to SF_6 during the etching process. Process pressure was adjusted by APC from 30% to 75% to understand the process pressure effect on RIE lag. The plasma source was generated by a 600 W 13.56 MHz RF generator and biased by a 10–14 W 13.56 MHz RF generator at the platen during the etching step. The wafer was electrostatically clamped and cooled by backside helium flow. A scanning electron microscope (SEM) was used to examine the etching result after the ASE^{TM} process.

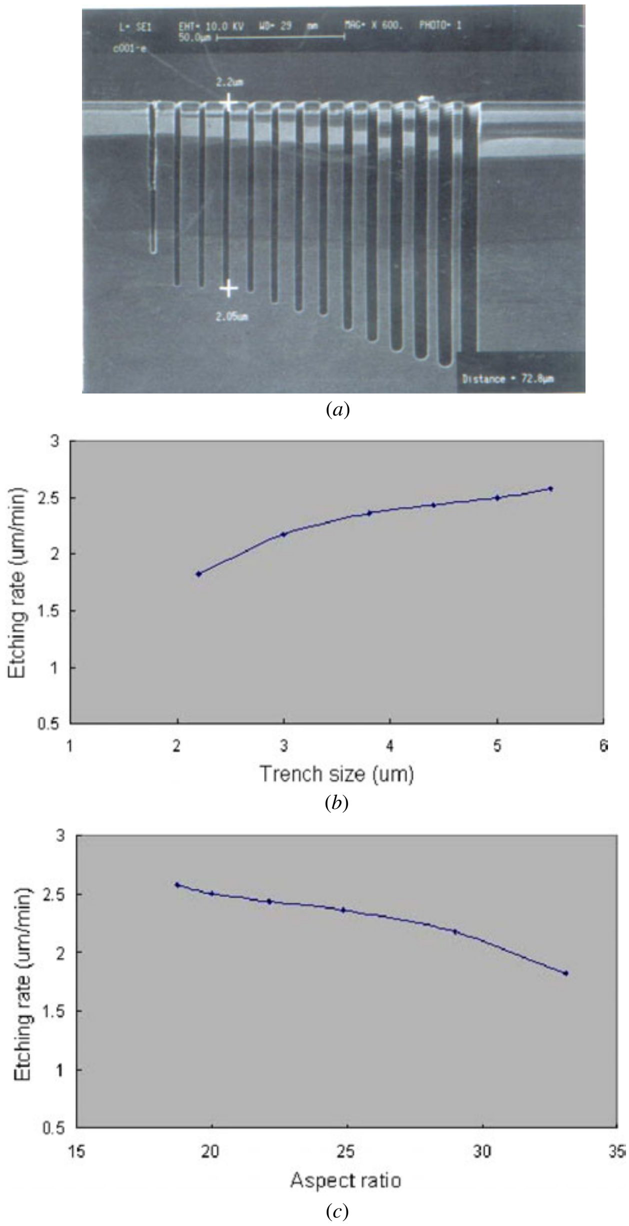
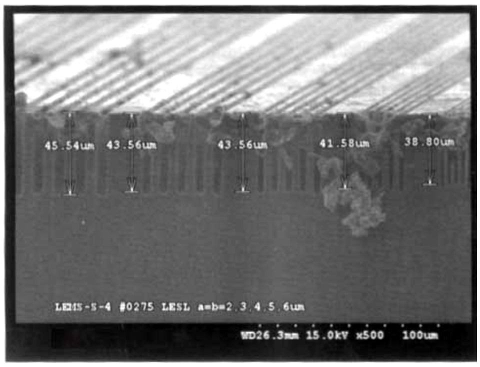


Figure 2. The high aspect ratio silicon trenches with nearly vertical sidewall profile obtained by inductively coupled plasma (ICP) etching: (*a*) SEM micrograph, (*b*) RIE lag and (*c*) aspect ratio dependent etching phenomena. The feature sizes of trenches from the fourth to ninth trenches from the left side are between 2.2 and 5.5 μm. The etch rates measured are from 1.82 to 2.58 μm min^{-1} while the aspect ratios obtained are from 33.1 to 18.7. The larger the trench size, the higher the etching rate and lower the aspect ratio.

3. Results and discussion

The inductively coupled plasma etching rate is related to the mask pattern geometry. Figure 2(*a*) shows the high aspect ratio silicon trenches with nearly vertical sidewall profile obtained by ICP etching. A RIE lag phenomenon occurs at such high aspect ratio trenches with constant length structure as shown in figure 2(*b*). The etching process is conducted by reactant gases in an etching step of 120 sccm, 40 mTorr SF_6 with 10% O_2 and a passivation step with 85 sccm, 22 mTorr C_4F_8



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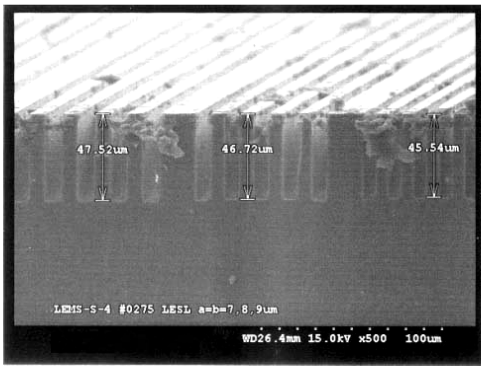

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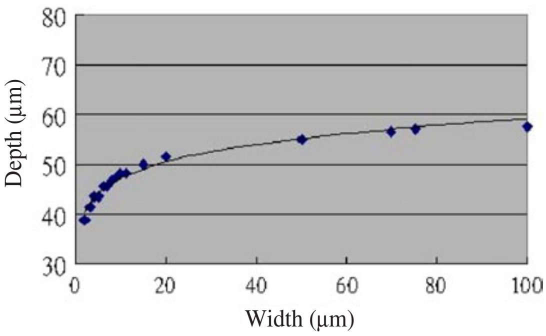
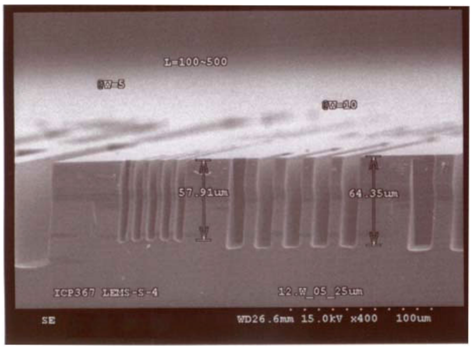


(*c*)

Figure 3. SEM micrographs of the etching depth of trenches with feature width: (*a*) 2 to 6 μm and (*b*) 7 to 9 μm in addition to (*c*) the plot of overall etching depth variation for a feature width of 2 to 100 μm at constant length of 2 mm, respectively. The etching time is 30 min. The RIE lag phenomenon becomes slower as the feature width increases.



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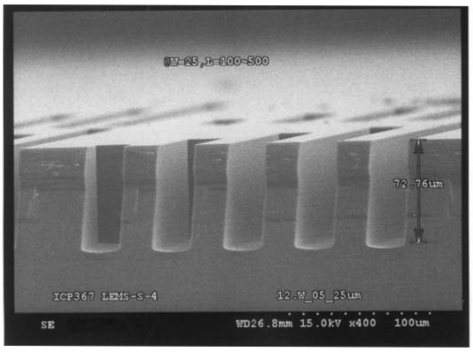

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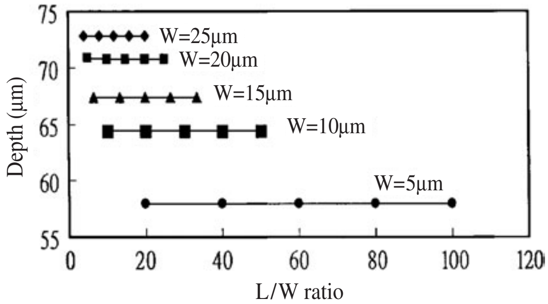


(*c*)

Figure 4. SEM micrographs of the etching depth of trenches with feature lengths from 100 to 500 μm at constant width: (*a*) 5–10 μm and (*b*) 25 μm in addition to (*c*) the plot of overall etching depth, respectively. The etching depth of trenches is nearly the same as the feature width which is fixed. But the etching depth varies as the feature width is changed. The feature width of pattern geometry is a dominating factor affecting the RIE lag in ICP etching.

for a reaction time of 40 min. The different feature sizes of trenches from the fourth to ninth trenches from the left side are to range from 2.2 to 5.5 μm. The etch rates measured are in a range between 1.82 and 2.58 μm min^{-1}. The aspect ratios obtained on these trenches are between values of 33.1 and 18.7. The larger the trench size, the higher the etching rate and the lower the aspect ratio. The aspect ratio dependent etching result is shown in figure 2(*c*) as expected due to RIE lag. The detailed feature width effect on the etching rate of trenches with constant length of 2 mm is shown in figures 3(*a*)–(*c*). The feature width of trenches ranges from 2 to 100 μm and etching time is 30 min. The etching depth varies from 38.80 to 47.52 μm for a feature width of 2 to 9 μm as shown in figures 3(*a*) and (*b*), respectively. The RIE lag phenomenon becomes slower as the feature width increases. The overall etching depth variation for a feature width of 2 to 100 μm is plotted in figure 3(*c*) whose trend is similar to figure 2(*b*). The feature width plays an important role in the RIE lag in ICP etching.

Figures 4(*a*)–(*c*) show the etching depth of trenches with feature lengths from 100 to 500 μm at constant width of 5–10 and 25 μm in addition to the plot of overall etching depth, respectively. The etching depth of trenches is nearly the same as the feature width which is fixed. But the etching depth varies as the feature width is changed. For example, the etching depth is 57.91 μm for the fixed feature width of 5 μm while the etching depth is 64.35 μm for the fixed feature width of 10 μm as shown in figure 4(*a*), and the etching depth is 72.76 μm for the fixed feature width of 25 μm as shown in

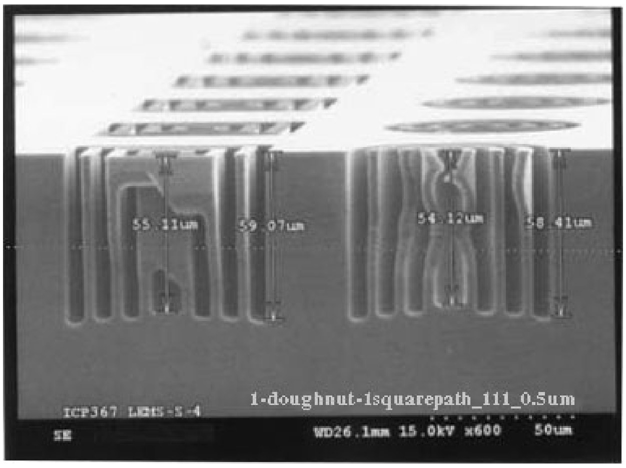
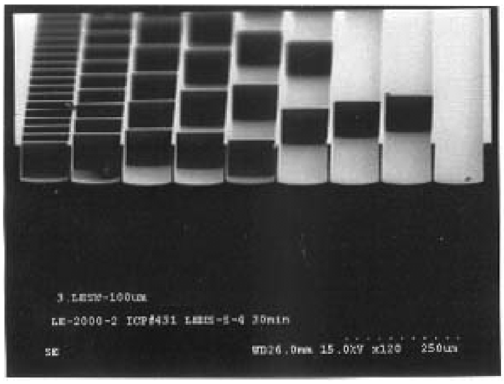


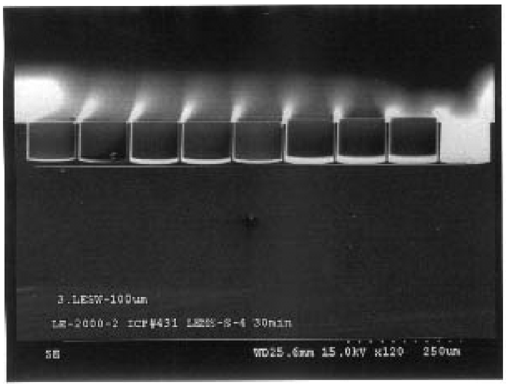
Figure 5. The etching depth of rectangular and circular doughnut trenches with constant width or radius of 5 μm. The etching depths of rectangular doughnut trenches on the left increase from 55.11 μm inner to 59.07 μm outer with larger continuous feature area while the etching depths of circular doughnut trenches on the right increase from 54.12 μm inner to 58.41 μm outer. The larger the continuous feature area, the deeper the etching depth. The rectangular doughnut has a larger area than the circular doughnut, which leads to the higher etching rate or depth.

figure 4(*b*). The relationship among etching depth, width and L/W ratio of trench is drawn in figure 4(*c*). The etching depth of a rectangular trench increases with increasing feature width and is insensitive to the feature length. It indicates that the feature width of pattern geometry is a dominating factor affecting the RIE lag in ICP etching. Besides the feature width, the continuous feature area or pattern shape also affects the etching rate. Figure 5 shows the etching depth of the doughnut trench structure with constant width or radius of 5 μm. The etching depth of the rectangular doughnut trench on the left of figure 5 increases from 55.11 μm inner to 59.07 μm outer with larger continuous feature area while the etching depth of the circular doughnut trench on the right of figure 5 increases from 54.12 μm inner to 58.41 μm outer. The larger the continuous feature area, the deeper the etching depth. In addition, the rectangular doughnut has a larger area than the circular doughnut, which leads to the higher etching rate or depth. Etching rates of the doughnut trenches are sensitive to feature area in addition to width. This implies that the feature area of pattern geometry is a secondary factor affecting the RIE lag in ICP etching.

To understand how feature area affects the RIE lag, we designed trenches at fixed width of 100 μm and varied length from 50 to 10 000 μm with respect to length-to-width (L/W) ratio of 0.5 to 100. Figures 6(*a*)–(*c*) show the SEM micrographs of tilt and front view of such patterns in addition to a plot of the etching results. The etching depth increases with increasing L/W ratio initially and then becomes constant at larger L/W ratio. This is in agreement with the results in figures 4(*a*)–(*c*). If the L/W ratio is large enough, the etching rate or depth of trenches at constant width is insensitive to the feature length corresponding to the L/W ratio or feature area. But the etching depth is affected by the feature length or area as the L/W ratio decreases. The smaller the feature length or area, the shallower the etching depth. It also agrees with



(*a*)



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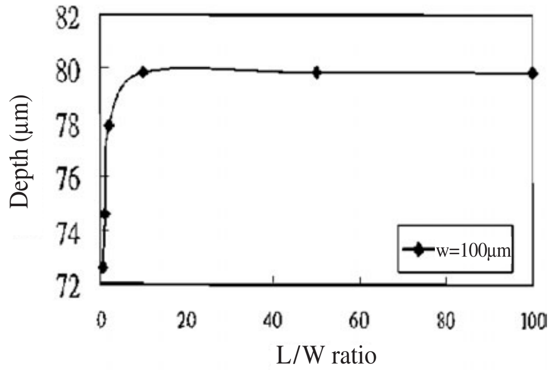
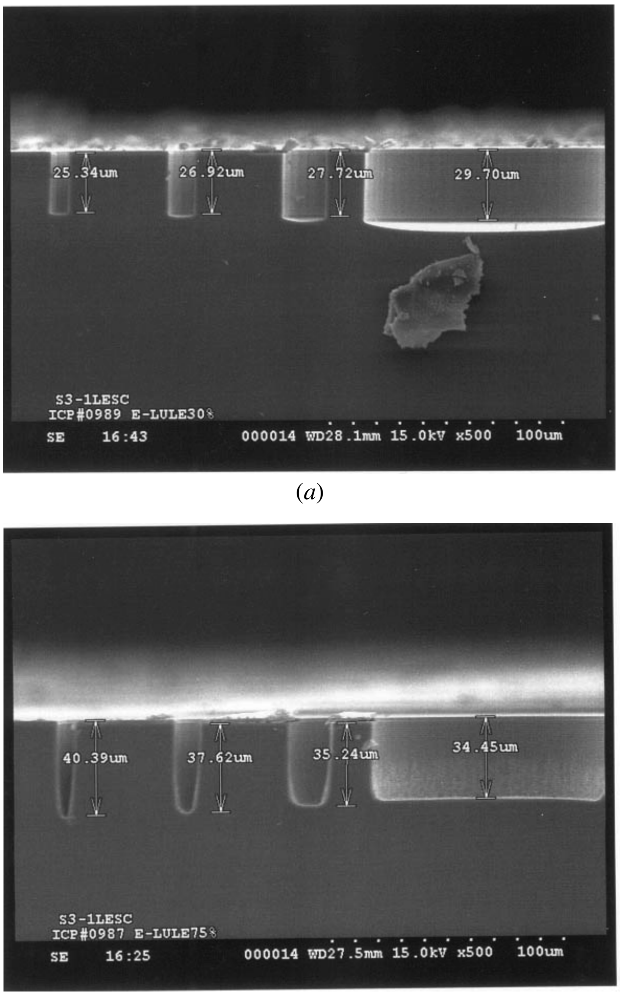


(*c*)

Figure 6. SEM micrographs of the etching depth of trenches at fixed width of 100 μm and varied length from 50 to 10 000 μm with respect to L/W ratio of 0.5 to 100: (*a*) tilt and (*b*) front view in addition to (*c*) the plot of etching result. The etching depth increases with increasing L/W ratio initially and then becomes constant at larger L/W ratio.

the result in figure 5 that feature area is a secondary factor affecting the RIE lag. A transition point exists between the feature length-sensitive and feature length-insensitive area in relation to the etching rate of trenches at constant width and is related to the feature dimensions.

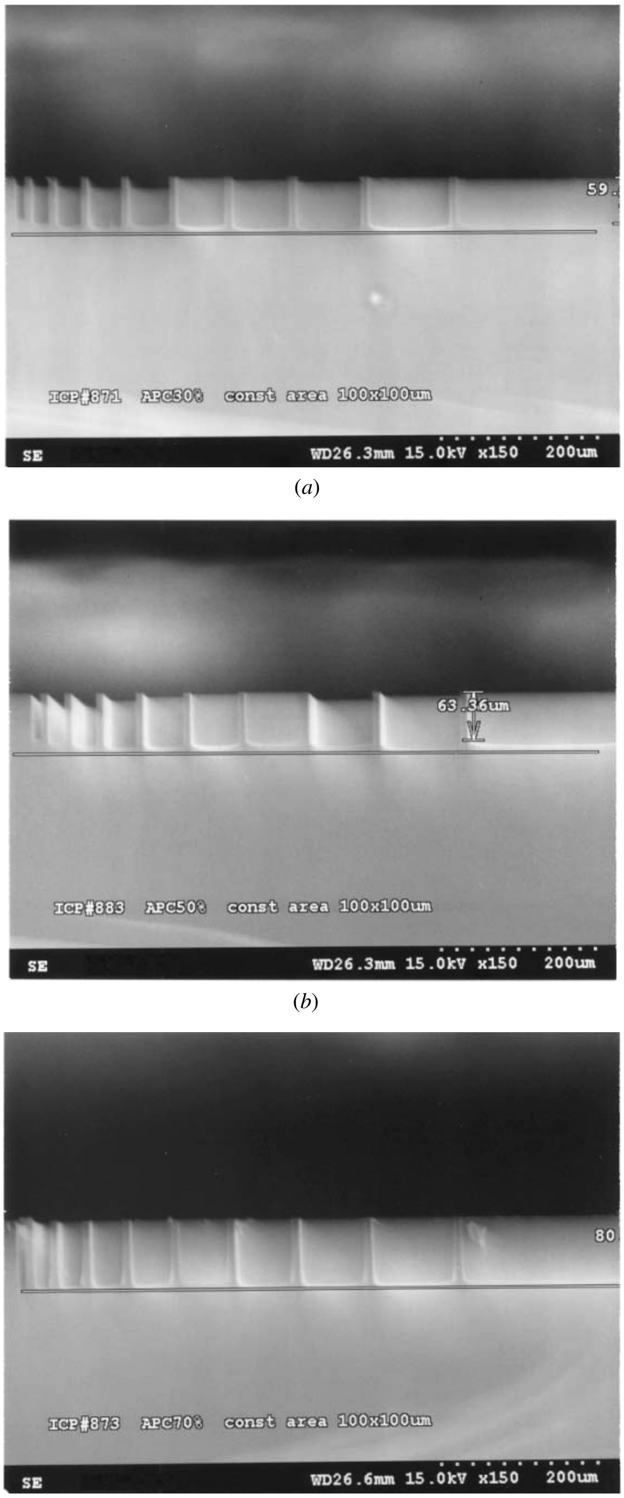
The ICP etching behaviour is dependent on the pattern geometry and process conditions. To study the process parameter effect on RIE lag in ICP etching, we fixed the secondary factor of feature area to decrease the extra geometry contribution to RIE lag. The process pressure is a crucial factor in the RIE lag control. Figures 7(*a*) and (*b*) show



(*a*)

(*b*)

Figure 7. SEM micrographs of the etched doughnut trench with constant area of $\pi \times 52.5^2$ μm^2 at pressures (*a*) APC 30% and (*b*) APC 75%. The RIE lag exists at a low pressure of APC 30% while an inverse RIE lag appears at a much higher pressure of APC 75%.



(*a*)

(*b*)

(*c*)

Figure 8. SEM micrographs of etched rectangular trench structures with constant area of 10 000 μm^2 at pressures (*a*) APC 30%, (*b*) APC 50% and (*c*) APC 70%. The width is varied from 10 to 100 μm while length is varied from 1000 to 100 μm. Obvious RIE lag is present at APC 30% and reduced at 50%, and RIE lag-eliminated trenches are obtained at APC 70%.

SEM micrographs of etching the doughnut trench structure with constant area of $\pi \times 52.5^2$ μm^2 at a pressure of APC 30% (20 mTorr (etching)/10 mTorr (passivation)) and APC 75% (54 mTorr (etching)/32 mTorr (passivation)) for 30 min, respectively. The RIE lag exists at a low pressure of APC 30% while an inverse RIE lag appears at a high pressure of APC 75%. The inverse RIE lag means the higher etching rate occurs at smaller feature width. It implies that another mechanism exists in Si DRIE with respect to the previous reports [9–12]. A possible new reaction mechanism is C_xF_y radicals dissociated from C_4F_8 passivation gas playing an important role in affecting the etching rate ratio of different feature sizes. The total etching rate is related to three steps: (1) the formation of C_4F_8 passivation film on the Si trench by C_xF_y radicals, (2) the removal of C_4F_8 passivation film at the trench bottom by S_xF_y ions dissociated from SF_6 etching gas and (3) the arrival of F radicals dissociated from SF_6 etching gas to etch the Si surface at the trench bottom. The formation and removal of C_4F_8 passivation film impedes the increase of etching depth while the arrival of SF_6 etching gas enhances the etching process. These two contrary reaction mechanisms compete by density of radicals and ions dissociated from C_4F_8 and SF_6 reaction gases and affect the total etching

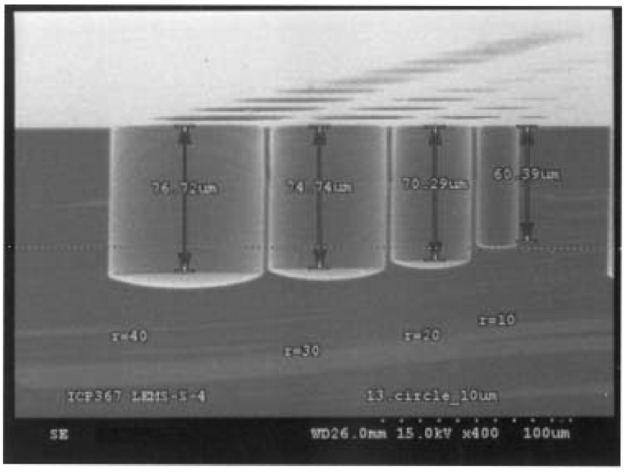
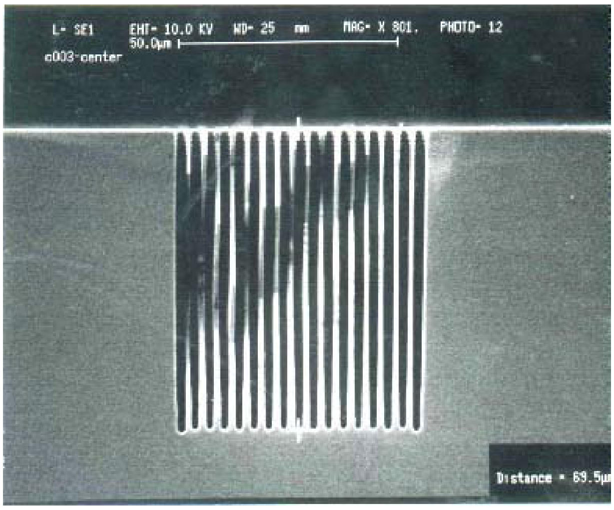


Figure 9. SEM micrograph of the holes with radius from 10 to 40 μm after 30 min etching. The etching depth increases from 60.39 to 76.72 μm and RIE lag reduces with increasing radius.

rate. The aspect ratio will also affect the formation and removal of C_4F_8 passivation film at different feature width due to the mass transport and/or scattering of C_xF_y radicals density and the S_xF_y ion density. A smaller width with larger aspect ratio will lead to thinner passivation film than a larger width. The difference of passivation film thickness will become larger at higher pressure because the increasing pressure will proportionally increase the C_xF_y and F radical density and decrease the S_xF_y ion density [15]. Although the removal of passivation film by ions under bias will decrease with aspect ratio due to scattering, the effect on different features will be small at the low ion density of high pressure and improved by higher bias to increase their directionality. So Si etching dominated by F radical density, which is nearly independent of aspect ratio [16], will start earlier in a small width feature than in a large one. It becomes more obvious at higher pressure due to the increase of C_xF_y and F radical density and decrease of S_xF_y ion density [15]. As the pressure increases, the increasing C_xF_y radical density and reducing S_xF_y ion density will delay longer the start of etching of the Si surface at the bottom of the large feature than of the small one. The etching rate is enhanced more at the small feature width than at the large one. This will lead to the reduction of RIE lag. If the pressure is very high, the inverse RIE lag occurs. It indicates that the cause of RIE lag in ICP etching is primarily attributed to the formation and removal of passivation film at the bottom of the trench, together with feature geometry. Since RIE lag and inverse RIE lag have been achieved, we could find an optimum condition to obtain the RIE lag-eliminated feature. Figures 8(*a*)–(*c*) show the SEM micrographs of etched trench structures at pressures of APC 30%, 50% and 70%, respectively. The patterns are rectangles with constant area of 10 000 μm^2. The width is varied from 10 to 100 μm while length is varied from 1000 to 100 μm. Obvious RIE lag is present at APC 30% and reduces at higher pressure of APC 50%. Finally, the RIE lag-eliminated trench is obtained at APC 70%. It is an interesting phenomenon. The etching rate could be adjusted by the pattern geometry and process parameters. Different structures with RIE lag, RIE lag-elimination or inverse RIE lag could be fabricated by proper pattern and process design in future.



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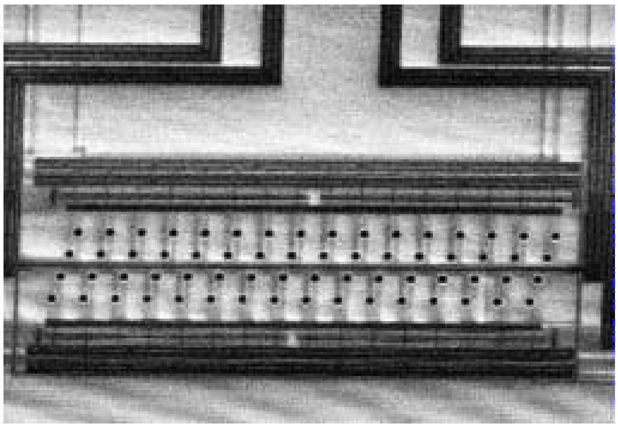

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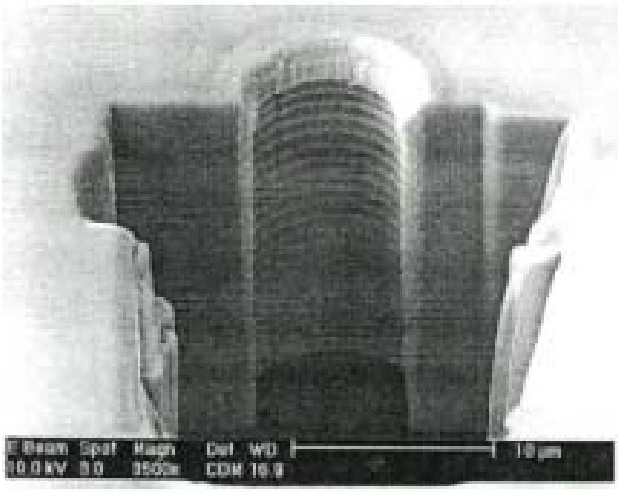

(*c*)

Figure 10. SEM micrographs of (*a*) high aspect ratio microstructure of holes with a nominal diameter of 1.5 μm and nearly the same etching depth of 69.5 μm, (*b*) plane-view nozzle array with circuit layout and (*c*) cross section of one nozzle.

In the application of an etched hole whose L/W ratio is 1, the etching depth is related to the diameter of the hole. Figure 9 shows etched holes with radius from 10 to 40 μm after 30 min etching at APC 70%. The etching depth increases from 60.39 to 76.72 μm and RIE lag reduces with increasing radius. This is in agreement with the results of figures 3(*c*) and 4(*c*). In general, holes with the same diameter are preferred for the fabrication of high aspect ratio microstructures (figure 10(*a*)) and nozzle arrays (figures 10(*b*)–(*c*)). Figure 10(*a*) shows the nominal hole diameter of 1.5 μm and nearly the same etching depth of 69.5 μm. The aspect ratio is very high, up to 46. Figures 10(*b*)–(*c*) show SEM micrographs of a plane-view of a nozzle array with circuit layout and a cross section of one nozzle, respectively. The nominal nozzle diameter is 10 μm and the profile is vertical and with scallop rings on it due to the ASE^{TM} switching process. This nozzle array is potentially useful for inkjet application in microfluidics.

4. Conclusions

Three major kinds of patterns of constant length, constant width and constant area with shapes of rectangles, squares, circles and doughnuts are designed to understand the factor of pattern geometry affecting RIE lag in ICP etching. A RIE lag phenomenon occurs in many patterns and becomes more severe at smaller feature width and higher etching depth. In general, the smaller the feature size, the lower the etching rate and the more obvious the RIE lag. Experimental results reveal that the primary dominating factor is feature width and secondary factors are feature area and L/W ratio. A transition point exists between the feature length-sensitive and length-insensitive areas of the etching rate of trenches at constant width and is related to the feature dimensions. A RIE lag elimination for different feature size at constant area can be achieved by a higher process pressure of APC 70%. The inverse RIE lag phenomenon appears at a much higher pressure of APC 75%. This will lead to three stages of RIE lag transition: reduced RIE lag, lag elimination and inverse RIE lag. It implies that a possible new reaction mechanism is C_xF_y radicals dissociated from C_4F_8 passivation gas playing an important role in affecting the etching rate ratio of different feature sizes. The formation and removal of C_4F_8 passivation film impedes the increase of etching depth while the arrival of SF_6 etching gas enhances the etching process. As the pressure increases, the increasing C_xF_y radical density for the formation of passivation film and reducing S_xF_y ion density to remove passivation film, will delay the start of etching of the Si surface at the bottom in a large feature longer than in a small one. The etching rate is enhanced more at a small width than at a large one. This will lead to the reduction of RIE lag. If the pressure is very high, inverse RIE lag occurs. It indicates that the effect of C_4F_8 passivation gas on RIE lag is very important in Si deep etching. It is an interesting phenomenon. The etching rate could be adjusted by the pattern geometry and process parameter. Different structures with RIE lag, RIE lag-elimination or inverse RIE lag could be fabricated by proper pattern and process design in future. Holes with the same diameter for the fabrication of high aspect ratio microstructures and nozzle arrays have been demonstrated.

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